DATA ANALYTICS
Unit 2: Regression Analysis
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Lecture 26: MLR – Validation

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Data Analytics

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Slides excerpted from: U. Dinesh Kumar, "Business Analytics", Wiley, 2nd Edition 2022

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In this session

- Validation of the MLR model
  1. Co-efficient of multiple determination (R^2) and Adjusted R^2
  2. Hypothesis test for individual variables (t-test)
  3. Analysis of variance for overall model validity (F-test)
  4. Residual analysis
  5. Presence of Multi-collinearity
  6. Check for auto-correlation
  7. Outlier analysis
     - Distance measures to find influential points

1. Coefficient of Multiple Determination (R^2) and Adjusted R^2

- In case of MLR, SSE will decrease as the number of explanatory variables increases, and SST remains constant
- Recall that R-square is defined as

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- So, it is possible, that R-square will increase even when there is no statistically significant relationship between the explanatory variable and the response variable
- To counter this, R^2 value is adjusted by normalizing both SSE and SST with the corresponding degrees of freedom
- The adjusted R-square with k predictors is given by

$$R^2 = 1 - \frac{SSE/(N-k-1)}{SST/(N-1)}$$

1. Coefficient of Multiple Determination (R^2) and Adjusted R^2 (cont.)

- To understand how degrees of freedom work, let us take the example of an equation with 3 unknowns

$$x_1 + x_2 + x_3 = 500$$

freedom to choose

freedom to choose

always determined as $500-(x_1+x_2)$

- For an equation with k variables, we have the freedom to choose k-1 of them
- In terms of regression, for a dataset with n observations and k predictors (k+1 partial regression coefficients), the number of values we can freely choose is **n-k-1**
  - A good explanation can be found here
- While R-square is a non-decreasing function, adjusted R-square is not
- The adjusted R-square value is always less than or equal to the R-square value
- No increase in adjusted R-square after adding a new predictor variable to the model may indicate that the newly added variable may not be statistically significant or it is not explaining the variation in the response variables that is not explained by the variables that are already present in the model

2. Hypothesis test for Individual Variables in MLR – t-test

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- Checking the statistical significance of individual variables is achieved through t-test
- The estimate of regression coefficient is given by

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

- The estimated value of regression coefficient is a linear function of the response variable Y
- Since we assume that the residuals follow normal distribution, Y follows a normal distribution, and the estimate of regression coefficient also follows a normal distribution
- Since the standard deviation of the regression coefficient is estimated from the sample, we use a t-test
- The null and alternative hypotheses in the case of an individual independent variable X_i and the dependent variable Y is given, respectively, by

$H_0 : \beta_i = 0$ (no relationship b/w X_i and Y)

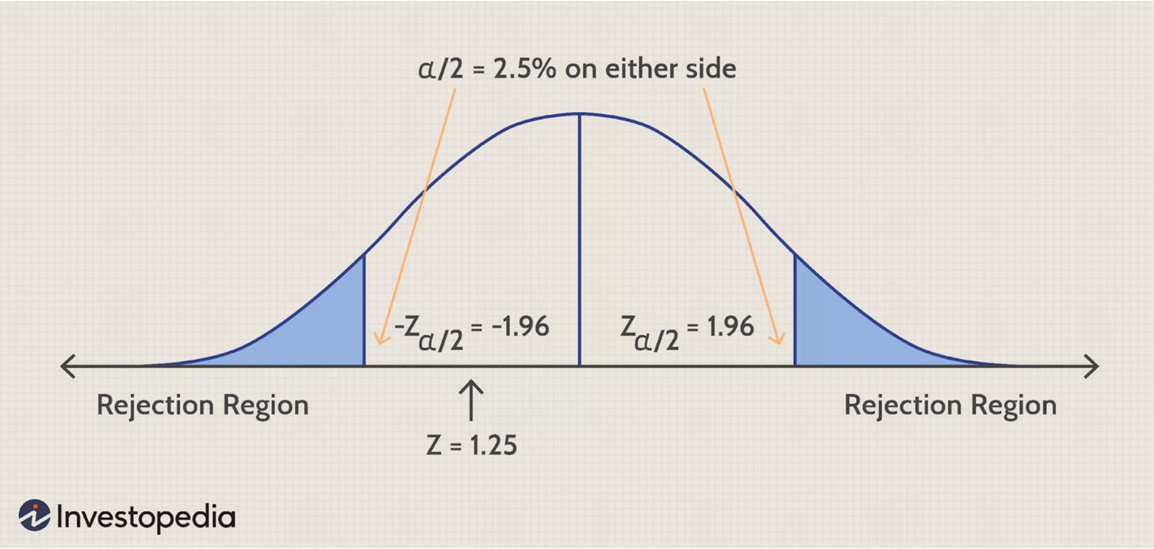
$H_a : \beta_i \neq 0$ (relationship b/w X_i and Y)

2. Hypothesis test for Individual Variables in MLR – t-test (cont.)

- Hypothesis test (two-tailed test)
  - $H_0: \beta_1 = 0$ (no linear relationship between X and Y)
  - $H_1: \beta_1 \neq 0$ (linear relationship between X and Y)

- t-statistic

$$t = \frac{\hat{\beta}_i - 0}{S_e(\hat{\beta}_i)} = \frac{\hat{\beta}_i}{S_e(\hat{\beta}_i)}$$



Two-tailed test:
source

3. Analysis of Variance for Overall Model Validity – F-test

- Analysis of Variance (ANOVA) is used to validate the overall regression model
- If there are k independent variables in the model, then the null and the alternative hypotheses are

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$$
$$H_a: \text{not all } \beta_i\text{'s are } 0$$

- Note that the statement in alternative hypothesis is that “not all betas are zero”, that is, some of those beta values may be zero
- That is the reason why we have to do the t-test to check the existence of statistically significant relationship between individual explanatory variables and the response variable
- F-statistic is given by (notice the degrees of freedom)

$$F = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(N-k-1)}$$

3. Analysis of Portions of a Model – Partial F-test

- In many cases, data scientists may like to validate the portions of the model or a subset of explanatory variables
- Assume that the data set has N observations (sample size), and we define two models named **full model** which has **k** independent variables and **reduced model** which has **r** independent variables (**r** < **k**) defined as follows

Full model (all k variables)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

Reduced model (r variables, r < k)

$$Y = \alpha_0 + \alpha_1 X_1 + \alpha_2 X_2 + \cdots + \alpha_r X_r$$

- Objective of partial F-test: check where the additional variables (X_{r+1}, X_{r+2}, ..., X_k) in the full model are statistically significant

3. Analysis of Portions of a Model – Partial F-test (cont.)

- The corresponding partial F-test has the following null and alternative hypotheses

$$H_0: \beta_{r+1} = \beta_{r+2} = \cdots = \beta_k = 0$$

$$H_a: \text{not all } \beta_{r+1}, \ldots, \beta_k \text{ are } 0$$

- Partial F statistic

reduced model

full model

$$\text{Partial } F = \left(\frac{(SSE_R - SSE_F)/(k-r)}{MSE_F} \right)$$

full model with N-k-1 dof

4. Residual Analysis

- Residual analysis is important for checking assumptions about normal distribution of residuals, homoscedasticity, and the functional form of a regression model

- If the residuals do not follow normal distribution, then we cannot trust the p-values of t-test and F-test since for the statistic to follow t-distribution and F-distribution, the residuals should follow normal or approximate normal distribution

- There are many reasons why residuals may not be normal; one such case is misspecification of functional form of regression, that is, the data scientist may have used linear model instead of log-linear or log-log model

- Similar to residual analysis of SLR
  - Construct P-P plots

References

- Business Analytics by U. Dinesh Kumar – Wiley 2nd Edition, 2022 Chapter 10

- U. Dinesh Kumar’s slides

- R-Squared, Adjusted R-Squared and the Degree of Freedom, Medium: https://medium.com/geekculture/r-squared-adjusted-r-squared-and-the-degree-of-freedom-80e7203a7e27

THANK YOU

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